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PACKING BORE WEAR SLEEVE RETAINER SYSTEM

Abstract

A fluid end includes a housing having a bore extending toward a cavity and a wear sleeve positioned within the bore. The fluid end also includes a plunger positioned within a plunger bore extending through the wear sleeve, the plunger reciprocating within the plunger bore. The fluid end further includes a wear sleeve retainer coupled to the housing and positioned to block axial movement of the wear sleeve, the wear sleeve retainer having external threads along a body that engage internal threads formed in the housing. The fluid end also includes an anti-rotation system, coupled to the housing, the anti-rotation system engaging the wear sleeve retainer to block rotation of the wear sleeve retainer in at least one direction. The fluid end further includes a packing nut coupled to the wear sleeve retainer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. Non-Provisional application Ser. No. 18/771,874, filed Jul. 12, 2024, titled “PACKING BORE WEAR SLEEVE RETAINER SYSTEM,” now U.S. Pat. No. 12,270,394, issued Apr. 8, 2025, which is a continuation of U.S. Non-Provisional application Ser. No. 17/890,975, filed Aug. 18, 2022, titled “PACKING BORE WEAR SLEEVE RETAINER SYSTEM,” now U.S. Pat. No. 12,049,889, issued Jul. 30, 2024, which is a continuation-in-part of U.S. Non-Provisional application Ser. No. 16/916,593, filed Jun. 30, 2020, titled “PACKING BORE WEAR SLEEVE RETAINER SYSTEM,” now U.S. Pat. No. 11,421,680, issued Aug. 23, 2022, the disclosures of which are incorporated herein by reference in their entireties for all purposes.

TECHNICAL FIELD

[0002] Embodiments of the subject matter disclosed herein generally relate to pump systems, and in particular to retainer systems.

BACKGROUND

[0003] Pumping systems may be used in a variety of applications, especially industrial applications where pumping systems are used to elevate a working fluid pressure. One such application is hydraulic fracturing systems, which use high pressure pumps to increase a fluid pressure of a working fluid (e.g., fracturing fluid, slurry, etc.) for injection into an underground formation. The working fluid may include particulates, which are injected into fissures of the formation. When the fluid is removed from the formation, the particulates remain and “prop” open the fissures, facilitating flow of oil and gas. The abrasive fluid, along with the high operating pressures, may lead to erosion within the pumping system, which may affect different components in different ways. Traditional systems have either modified materials of construction, for example by using expensive metals or coatings, or by introducing sacrificial components that are replaced over time. One such component is a wear sleeve, which is a cylindrical sacrificial component that lines a fluid end packing bore to provide a sealing surface for various sealing elements, such as plunger packing. The wear sleeve is designed to be sacrificial and a sealing surface that degrades over time, and as a result, leaks may occur. When the sealing surface degrades, the wear sleeve is removed and replaced. Typically, these wear sleeves are bolted directly onto the pump body, which uses multiple threaded fittings and may also be difficult to access.

SUMMARY

[0004] Applicant recognized the problems noted above herein and conceived and developed embodiments of systems and methods, according to the present disclosure, for retainer systems.

[0005] In an embodiment, a wear sleeve retainer system includes a wear sleeve retainer and an anti-rotation system. The wear sleeve retainer includes a body portion extending for a length, the body portion having a bore extending along an axis. The wear sleeve retainer also includes a mating component extending from the body portion and into the bore, the mating component being annular

and having a smaller diameter than a bore diameter. The wear sleeve retainer further includes threads arranged circumferentially about at least a portion of an outer diameter of the body portion. The wear sleeve retainer includes a profile formed about at least a portion of the body portion, the profile including a plurality of receptacles. The anti-rotation system includes a locking fastener adapted to engage a receptacle of the plurality of receptacles.

[0006] In an embodiment, a fluid end, includes a housing having a bore extending toward a cavity, a wear sleeve positioned within the bore, a plunger positioned within a plunger bore extending through the wear sleeve, the plunger reciprocating within the plunger bore, and a wear sleeve retainer coupled to the housing and positioned to block axial movement of the wear sleeve, the wear sleeve retainer having external threads along a body that engage internal threads formed in the housing. The fluid end also includes an anti-rotation system, coupled to the housing, the anti-rotation system engaging the wear sleeve retainer to block rotation of the wear sleeve retainer in at least one direction and a packing nut coupled to the wear sleeve retainer.

[0007] In an embodiment, a method for installing a retainer system includes positioning packing seal elements within a diameter of a wear sleeve while the wear sleeve is external to a fluid end housing. The method also includes positioning the wear sleeve within a bore formed in the fluid end housing. The method further includes securing a wear sleeve retainer to the fluid end housing by threading the wear sleeve retainer into the fluid end housing, wherein external threads on a body of the wear sleeve retainer engage internal threads in the bore, the wear sleeve retainer engaging at least a portion of the wear sleeve. The method includes securing an anti-rotation system to the wear sleeve retainer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The present technology will be better understood on reading the following detailed description of non-limiting embodiments thereof, and on examining the accompanying drawings, in which:

[0009] FIG. 1 is a perspective view of an embodiment of a fluid end, in accordance with embodiments of the present disclosure;

[0010] FIG. 2 is a perspective view of an embodiment of a fluid end, in accordance with embodiments of the present disclosure;

[0011] FIG. 3 is a perspective view of an embodiment of a retainer system, in accordance with embodiments of the present disclosure;

[0012] FIG. 4 is a perspective view of an embodiment of a retainer system, in accordance with embodiments of the present disclosure;

[0013] FIG. 5 is a cross-sectional view of an embodiment of a retainer system for a plunger assembly, in accordance with embodiments of the present disclosure;

[0014] FIG. 6 is an exploded view of an embodiment of a retainer system, in accordance with embodiments of the present disclosure;

[0015] FIG. 7 is a perspective view of an embodiment of a wear sleeve retainer, in accordance with embodiments of the present disclosure;

[0016] FIG. 8 is a perspective view of an embodiment of a lock body, in accordance with embodiments of the present disclosure;

[0017] FIG. 9 is a front view of an embodiment of a lock body, in accordance with embodiments of the present disclosure;

[0018] FIG. 10 is a side view of an embodiment of a lock body, in accordance with embodiments of the present disclosure;

[0019] FIGS. 11A-11E are perspective views of embodiments of an installation procedure, in

accordance with embodiments of the present disclosure;
[0020] FIGS. **12A** and **12B** are perspective views of embodiments of a ratchet and pawl anti-rotation system, in accordance with embodiments of the present disclosure;
[0021] FIG. **13** is a flow chart of an embodiment of a method for a retainer system, in accordance with embodiments of the present disclosure;
[0022] FIG. **14A** is a front view of an embodiment of an anti-rotation system, in accordance with embodiments of the present disclosure;
[0023] FIG. **14B** is a partial perspective view of an embodiment of an anti-rotation system, in accordance with embodiments of the present disclosure;
[0024] FIG. **14C** is a cross-sectional view of an embodiment of an anti-rotation system, in accordance with embodiments of the present disclosure; and
[0025] FIG. **15** is a perspective view of an embodiment of an anti-rotation system, in accordance with embodiments of the present disclosure.

DETAILED DESCRIPTION

[0026] The foregoing aspects, features, and advantages of the present disclosure will be further appreciated when considered with reference to the following description of embodiments and accompanying drawings. In describing the embodiments of the disclosure illustrated in the appended drawings, specific terminology will be used for the sake of clarity. However, the disclosure is not intended to be limited to the specific terms used, and it is to be understood that each specific term includes equivalents that operate in a similar manner to accomplish a similar purpose.

[0027] When introducing elements of various embodiments of the present disclosure, the articles “a”, “an”, “the”, and “said” are intended to mean that there are one or more of the elements. The terms “comprising”, “including”, and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Any examples of operating parameters and/or environmental conditions are not exclusive of other parameters/conditions of the disclosed embodiments. Additionally, it should be understood that references to “one embodiment”, “an embodiment”, “certain embodiments”, or “other embodiments” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Furthermore, reference to terms such as “above”, “below”, “upper”, “lower”, “side”, “front”, “back”, or other terms regarding orientation or direction are made with reference to the illustrated embodiments and are not intended to be limiting or exclude other orientations or directions.

[0028] Embodiments of the present disclosure include a packing bore wear sleeve retainer system with an integrated anti-rotation device for use on positive displacement reciprocating pumps. Such pumps include hydraulic fracturing pumps, mud pumps, and similar plunger or piston pumps. Embodiments incorporate a wear sleeve retainer that has an outer diameter that is threaded and an internal diameter that is threaded at an end for receiving the mating component packing nut. On the opposite end, the wear sleeve retainer is configured geometrically to capture and secure the wear sleeve when it is fully threaded into the packing bore of the fluid end. The sleeve retainer is designed to secure the wear sleeve into the gland of the fluid end bore tightly and incorporates sealing elements to prevent high-pressure fluid leakage from the pumping chamber along with preventing leakage of high-pressure grease or oil, which is used to lubricate the plunger.

[0029] Embodiments of the present disclosure include features that allow the wear sleeve retainer to be removed and reassembled into the fluid end quickly by the operator and with ease by the use of a separate spanner wrench attachment tool that can be used in conjunction with industry standard ratchet style wrenches. Furthermore, embodiments include an integrated anti-rotation device. When the wear sleeve retainer is fully threaded into the packing bore of the fluid end, the device set screw bolt is tightened down and secures into one of multiple slots that are positioned at multiple points 360 degrees around the outer diameter, preventing the wear sleeve retainer from unthreading from

the fluid end while in operation due to vibrations.

[0030] The wear sleeve is presented in various embodiments as a cylindrical sacrificial component that lines the fluid end packing bore to provide a sealing surface for the sealing elements (e.g., plunger packing), the wear sleeve is intended to be sacrificial, during pumping operations the packing will seal up on the plunger which is reciprocating in and out of the fluid end and the inner diameter of the wear sleeve. The sealing surface of the wear sleeve will begin to degrade with time due to high-pressure abrasive fluids eroding the material until it can no longer seal satisfactorily, and high-pressure fluid leakage will then occur causing washout metal erosion damage. At this point, maintenance occurs on the fluid end and the wear sleeve is removed from the fluid end and replaced with a new one. Embodiments of the present disclosure are directed toward a reliable, easy, and fast retention system to enable on-site maintenance by field service personnel. Prior art wear sleeve retainer systems incorporate a bolt on flange. The design is less reliable and slow to work on due to the many bolts that all required to be torqued down to specification. There is limited access space to maneuver tools to torque down the many bolts when the fluid end is attached to the pump, which is often the case when performing maintenance on the fluid end while the unit is on the job site. Another issue with prior art sleeve retainer designs is that they can at times begin to loosen or back out, causing the wear sleeve to be able to move back and forth in the gland, causing leakage or severe mechanical damage to the pump. Embodiments of the present disclosure overcome these problems by incorporating an integrated anti-rotation mechanical device that is fast and easy to use and prevents the wear sleeve retainer from backing out while pumping.

[0031] Embodiments of the present disclosure provide significant advantages over prior art systems and utilize a sleeve retainer system that threads into the fluid end via the use of a spanner wrench attachment and has an integrated anti-rotation lock mechanism to keep the sleeve retainer from backing out during pumping operations, thereby solving problems associated with back out during pumping, which may result in high pressure fluid leakage (washout erosion damage) or severe mechanical damage to pumping equipment. Embodiments of the present disclosure also eliminate the use of prior art bolts, which are torqued down to specification, this is difficult due to limited access behind the fluid end when being mounted to the power frame.

[0032] Embodiments of the present disclosure further incorporate one or more anti-rotation devices that may interact with a wear sleeve retainer positioned to maintain a position of a wear sleeve within a bore. A fastener may extend through at least a portion of a fluid end housing to engage a region of the wear sleeve retainer that is arranged within the bore. In at least one embodiment, a wedge-lock retainer profile may be incorporated into the system. In various embodiments, one or more steps or ridges of the wear sleeve retainer may include an aperture or orifice to receive the fastener. However, in other embodiments, the fastener may not interact with an aperture or orifice, which may increase useability of the system due to working in a variety of different alignments. The fastener may extend toward the wear sleeve retainer and be positioned proximate a ridge or step such that rotation of the ridge or step toward the fastener leads to the ridge or step engaging the fastener and thereby blocking rotation. The fastener may be arranged within a counterbored hole and extend through one or more of a top of the housing, a bottom of the housing, or a side of the housing. Additionally, the fastener may be positioned with a vertical orientation (e.g., axis of the fastener perpendicular to an axis of the bore) or at an orientation where the axis of the fastener is not perpendicular to the axis of the bore.

[0033] In at least one embodiment, systems and methods may incorporate a front-mounted system that engages the wear sleeve retainer to block rotation of the wear sleeve retainer. By way of example, a pad retainer may be incorporated into the system that is positioned to bear against a surface of the wear sleeve retainer, which may be textured or knurled surface to provide further resistance to rotation when incorporated with the pad retainer. The pad retainer may be fastened or otherwise secured to the housing and positioned to engage the textured surface of the wear sleeve retainer to apply a force against the wear sleeve retainer. In operation, as the wear sleeve retainer

begins to rotate and/or receives a force to drive rotation, the pad retainer may cause a frictional force on the surface to resist or overcome the rotational force, thereby blocking rotation of the wear sleeve retainer.

[0034] FIG. 1 is a perspective view of an embodiment of a fluid end **100**. As noted above, fluid ends **100** may be utilized in industrial applications, such as oil and gas applications, to deliver high-pressure fluids to piping components leading to wellbores. For example, hydraulic fracturing operations use fluid ends **100** to increase a fluid pressure for fracturing fluid, which may be corrosive and/or abrasive, prior to injection into a wellbore. Fluid ends **100** are often coupled to engines, which provide motive power to drive reciprocation of various plungers. The engines may provide sufficient power to pressure fluid to pressure ranges from 5,000-25,000 pounds per square inch (psi). It should be noted that these pressure ranges are provided by example and are not intended to limit the scope of the present disclosure, as the pressures ranges may be less than 5,000 psi or greater than 25,000 psi in various applications. Furthermore, as noted, while discussion may be directed toward hydraulic fracturing operations systems and methods may be used in a variety of other industries that incorporate pumping systems.

[0035] The illustrated fluid end **100** includes a manifold body or housing **102** that is illustrated as a single, unitary piece, but it should be appreciated that the housing **102** may be formed of multiple sections. As will be appreciated, the housing **102** may include a conduit or bore that is represented as an inlet that received a low pressure fluid, a chamber that receives the low pressure fluid, and an outlet that intersects the chamber and discharges the high pressure fluid.

[0036] In this embodiment, the fluid end **100** includes five different plunger assemblies **104**, each including a plunger **106**, a wear sleeve retainer **108**, a packing nut **110**, and a flange **112**. As shown, the wear sleeve retainer **108** is coupled directly to the housing **102** via fasteners **114**, which are bolts in the illustrated embodiment. Because space is at a premium at a well site, it may be difficult to access the fasteners **114** when the wear sleeve (not pictured) is replaced. This may be difficult or time consuming for operators, which may lead to increased downtime at the site. Embodiments of the present disclosure include an improved system for retaining the wear sleeve as well as components for installation and removal.

[0037] FIG. 2 is a perspective view of an embodiment of the fluid end **100** including the plunger assemblies **104** having a wear sleeve retainer system **200**. As will be described in detail below, the system **200** does not include the fasteners **114**, but rather, uses internal threads to couple components to the housing **102**. Such an arrangement enables faster assembly and disassembly, thereby reducing time for maintenance. Furthermore, embodiments may include an anti-rotation system **202** that reduces the likelihood of backing out of a wear sleeve retainer **204** that forms part of the system **200**. Accordingly, the wear sleeve (not pictured) is still replaceable and may be utilized as a sacrificial component, however, installation is simplified and faster.

[0038] FIG. 3 is a perspective view of an embodiment of the system **200** coupled to the housing **102**. In the illustrated embodiment, the wear sleeve retainer **204** is positioned against the housing **102**, for example, against a housing external face **300** (e.g., face). It should be appreciated that in other embodiments there may be a recessed portion in the housing face **300**, or a platform, to receive the wear sleeve retainer **204**. In other words, the housing external face **300** may not be planar in all embodiments. As will be described below, the wear sleeve retainer **204** may engage internal threads formed along a bore extending through the housing **102** that receives the plunger **106**. As a result, external fasteners for securing the wear sleeve retainer **204** to the housing **102** may be eliminated. That is, the fasteners **114** of FIG. 1 that provide a direct coupling that ends through the wear sleeve retainer **204** are reduced and/or eliminated by incorporating embodiments of the present disclosure.

[0039] The illustrated wear sleeve retainer **204** is secured against rotation by the anti-rotation system **202**, which includes a locking mechanism **302** and a locking fastener **304**. The locking mechanism **302** is secured to an aperture **306** formed in the housing **102**. In various embodiments,

the locking mechanism **302** is threaded into the aperture **306**, press fit into the aperture **306**, fastened to the aperture **306**, or the like. For example, in an embodiment, a threaded fitting may be utilized to secure the locking mechanism **302** to the housing **102** via the aperture **306**. In various embodiments, the aperture **306** is positioned in a particularly selected location to facilitate incorporation with the anti-rotation system and the wear sleeve retainer **204**, as will be described below. However, in various other embodiments, the locking mechanism **302** may be adjustable to enable a modification of an anti-rotation angle of **308** of the locking fastener **304**. As will be appreciated, even if threaded fasteners are utilized for the locking mechanism **302**, a total of five threaded fasteners would be used for the illustrated embodiment (e.g., one for each of the five plunger assemblies **104**), compared to potentially a dozen for each plunger assembly in prior configurations. The locking mechanism **302** receives the locking fastener **304**, which is illustrated as a threaded bolt, which extends toward and engages the wear sleeve retainer **204**. As shown, the locking fastener **304** is arranged at the anti-rotation angle **308** that is biased against a removal rotation for the wear sleeve retainer **204** (e.g., counter clockwise). By positioning the locking fastener **304** at the angle **308**, forces may be distributed along two force component directions (e.g., vertically and horizontally), thereby enabling smaller locking fasteners **304**. In this example, rotation in the counter clockwise direction is blocked due to engagement between the wear sleeve retainer **204** and the locking fastener **304**. Accordingly, problems with traditional systems associated with backing out of wear sleeve retainers secured directly to the housing **102** by fasteners are overcome because each of the fasteners, such as the fasteners **114** of FIG. 1, are subjected to forces along a single plane. Moreover, as will be described, the locking fastener **304** may be marked or otherwise used as an indicator during installation and/or maintenance procedures.

[0040] The illustrated wear sleeve retainer **204** includes an outer circumference profile **310** having a plurality of spaced apart valleys **312** separated by flats **314**. The valleys **312** are illustrated having a semi-circular shape with a radius, however, it should be appreciated that the valleys **312** may be any reasonable shape. For example, the valleys **312** may include sloped sides extending to trench or may include a single sloped side, among various other configurations. Moreover, the flats **314** may also be a different shape and are referred to as “flats” for illustrative purposes, but may include rounded edges or the like. In certain embodiments, the valleys **312** may also include a mating aperture for receiving the locking fastener **304**. Each of the valleys **312** extend for a longitudinal valley depth **316** toward the housing external face **300**. That is, the valleys **312** in the illustrated embodiment include a backstop **318**, which may be a portion of a threaded body portion that is installed within a bore formed in the housing **102**. The backstop **318** may provide a visual indication to the operator regarding installation of the wear sleeve retainer **204**. For example, the backstop **318** may be substantially flush with the housing external face **300** to indicate full installation. However, it should be appreciated that the backstop **318** may also be recessed relative to the housing external face **300** to provide room for the locking fastener **304**.

[0041] Further illustrated with respect to the wear sleeve retainer **204** are a plurality of blinds **320** positioned circumferentially about the wear sleeve retainer **204**. The illustrated blinds **320** are radially inward, with respect to the valleys **312**, and are positioned to align with the flats **314** in the illustrated embodiment. Such an arrangement is for illustrative purposes, and the blinds **320** may be particularly positioned based on a number of different factors. As will be described below, the blinds **320** may be utilized to receive a tool for installation of the wear sleeve retainer **204**. For example, an extrusion or extension of a tool may be fitted to engage one or more blinds **320** to enable rotation of the wear sleeve retainer **204**, thereby securing the wear sleeve retainer **204** to the housing **102**.

[0042] FIG. 3 also includes the packing nut **110** positioned outward of the wear sleeve retainer **204** and also the plunger **106** extending through aligned bores extending through the wear sleeve retainer **204** and the packing nut **110**. Accordingly, a familiar arrangement is maintained, which

may simplify installation procedures for operators. Moreover, the illustrated configuration may enable other equipment to be utilized in the course of traditional operations, such as packing nut locks bars and the like.

[0043] FIG. 4 is a front perspective view of an embodiment of the retainer system **200** in which the plunger **106** has been removed for clarity. As described above, in various embodiments the wear sleeve retainer **204** is installed within a bore **400** formed in the housing **102**. The bore may also include a wear sleeve, packing, and the like, as will be described in detail below. In this embodiment, the wear sleeve retainer **204** is installed along the bore **400** via external threads formed on the wear sleeve retainer **204** and internal threads of the bore **400**. Furthermore, in this embodiment, a plunger bore **402** is shown extending through both the wear sleeve retainer **204** and the packing nut **110**. In operation, the plunger **106** is installed through the plunger bore **402** and reciprocates back and forth, along a plunger bore axis **404** in order to pressurize fluid within the housing **102**.

[0044] As described above, the circumferential profile **310** is illustrated extending entirely around the wear sleeve retainer **204**, thereby enabling installation of the wear sleeve retainer **204** in any orientation that facilitates alignment with the threads. In this embodiment, each of the valleys **312** is equally spaced about an outer diameter **406** of the wear sleeve retainer **204**. However, it should be appreciated that different patterns or positions for the valleys **312** may be provided in various embodiments. Moreover, spacing between valleys **312** may be different. That is, a flat length **408** may vary at different regions. Furthermore, a valley length **410** may also vary between different valleys **312**. Accordingly, various profile **310** configurations may be particularly selected for different operational goals, such as reducing weight, driving alignment of components, and the like.

[0045] As noted above, the anti-rotation system **202** includes the locking mechanism **302** installed within the aperture **306** with the locking fastener **304** extending through the locking mechanism **302** at the angle **308**. This angle **308** may be particularly selected to transmit a rotational force in a counter clockwise direction, which would correspond to a direction that would unthread or back out the wear sleeve retainer **204**. The illustrated angle **308** is approximately 30 degrees. However, it should be appreciated that the angle **308** may be any reasonable angle to prevent rotation of the wear sleeve retainer **204**, such as approximately 15 degrees, approximately 35 degrees, approximately 45 degrees, approximately 50 degrees, or the like. Furthermore, it should be appreciated that the relative location of the anti-rotation system **200** is for illustrative purposes only and may be below the wear sleeve retainer **204**, next to the wear sleeve retainer **204**, or at any other reasonable location to engage at least a portion of the wear sleeve retainer **204** and to block rotation of the wear sleeve retainer **204**.

[0046] FIG. 5 is a cross-sectional view of an embodiment of the retainer system **200** coupled to the housing **102**. It should be appreciated that various features have been eliminated for simplicity with the following discussion. The illustrated embodiment includes the bore **400** extending through the housing **102** toward a cavity **500**. The bore **400** includes a first bore section **502**, a second bore section **504**, and a third bore section **506**, each having a different respective bore diameter. For example, a first bore section diameter **508** is larger than a second bore section diameter **510**, which is larger than a third bore section diameter **512**. It should be appreciated that this arrangement is for illustrative purposes and in various embodiments there may be more sections and/or different diameters.

[0047] The illustrated plunger assembly **104** includes a wear sleeve **514** arranged within the bore **400** and extending through each of the first, second, and third bore sections **502**, **504**, **506**. The wear sleeve **514** is a stepped sleeve having a transition **516** where the diameter changes. The illustrated wear sleeve **514** is positioned to bear against a wear sleeve seal **518** positioned within a wear seal groove **520** formed in the wear sleeve **514**. It should be appreciated that the seal **518** and groove **520** may also be formed in the housing **102** in other embodiments. The wear sleeve **514** also includes a shelf **522** that enables packing **524** to be installed along an inner portion of the wear

sleeve **514**, which may bear against the plunger **106** extending through the bore **400**. The wear sleeve **514** also includes an external seal **526** in an external seal groove **528** for engaging the wear sleeve retainer **514**. It should be appreciated that the external seal **526** and groove **528** may also be arranged within the wear sleeve retainer **204**.

[0048] In operation, the wear sleeve **514** is installed within the bore **400** and the wear sleeve retainer **204** is utilized to secure the wear sleeve **514** at a desired position. In this embodiment, the housing **104** includes threads **530**, which may engage mating threads **532** on the wear sleeve retainer **204**. The illustrated threads **530** are formed along the bore **400** at the first bore section **502**. In other words, the threads **530** in the illustrated embodiment may be described as being internal to the housing **102**. These threads **530** engage the mating threads **532** formed along a body outer circumference **534** of a body **536** of the wear sleeve retainer **204**. As will be appreciated, the body **536** may extend axially into the first bore section **502** a predetermined amount to facilitate engagement of the wear sleeve **514**. In this example, a mating component **538** extends radially inward, toward the axis **404**, to engage the wear sleeve **514**. As a result, axial movement of the wear sleeve **514** is blocked along the axis **404**. That is, axial movement in a first direction **540** toward the chamber **500** is blocked by a transition **542** along the bore **400** and axial movement in a second direction **544** toward the wear sleeve retainer **204** is blocked via the mating component **538** and an opposing force provided by the threads **530** and the mating threads **532**. Accordingly, the wear sleeve **514** is secured in position within the bore **400**.

[0049] In various embodiments, ports **546** are formed within the housing **102** and align with mating ports **548** formed in the wear sleeve retainer **204**. As a result, grease or other lubricants may be added to various components, such as the plunger **106**, without removing the packing assemblies **524**. Moreover, various seals may also be utilized to block fluid leakage, such as the external seal **526** and/or a wear sleeve seal **550** positioned in a wear sleeve seal groove **552** formed in the body **536**, which as noted above may also be formed in the housing **102**.

[0050] Installation may also include the packing nut **110**, which secures the packing **524** within the wear sleeve **514**. The packing nut **110** may couple to the wear sleeve retainer **204**, for example via threads **554** and mating threads **556** formed on the packing nut and the wear sleeve retainer **204**, respectively. However, it should be appreciated that other coupling devices, such as clamps or fasteners, may also be utilized. Accordingly, embodiments of the present disclosure provide the retainer system **200** for maintaining a position of the wear sleeve **514** within the bore **400** without using external threads to directly couple the wear sleeve retainer **204** to the housing **102**.

[0051] FIG. **6** is a partial exploded view of an embodiment of components of the retainer system **200** for securing the wear sleeve **514** within the bore **400**. As noted above, features have been eliminated for clarity and conciseness. The illustrated embodiment shows each of the wear sleeve **514**, wear sleeve retainer **204**, packing nut **110**, and plunger **106** being aligned along the axis **404**, thereby enabling coaxial alignment within the bore **400**.

[0052] FIG. **7** is cross-sectional perspective view of the wear sleeve retainer **204**. As noted above, the illustrated wear sleeve retainer **204** is a generally cylindrical component that includes a body **536** extending axially from a face end **700** that is substantially aligned with the external housing face **300** when installed within the housing **102**. The face end includes the blinds **320** arranged circumferentially about the face end **700**, as well as the profile **310** along the circumference. As noted above, the profile includes the valleys **312** and flats **314**, where the flats **314** extend for the length **408** and the valleys **312** extend for the length **410**, each of which may be adjusted as particularly selected for various applications. In various embodiments, the valleys **312** extend for the valley depth **316** that does not extend through an entire length **702** of the wear sleeve retainer **204**, but rather, to a backstop **318**. The backstop **318** abuts the threads **532** formed along the outer body diameter **534**, which may facilitate engagement with the threads **530** formed in the housing **102**.

[0053] The plunger bore **402** of the wear sleeve retainer **204** also includes the threads **556** for

coupling to the packing nut **110**. Also provided within the plunger bore **402** is the mating component **538**, which is illustrated as extending annularly around the plunger bore **402**. The mating component **538** engages the wear sleeve **514**, thereby blocking movement of the wear sleeve **514** out of the bore **400** toward the face end **700**.

[0054] FIG. **8** is a perspective view of an embodiment of a lock body **800** that may be utilized to install the wear sleeve retainer **204**. The illustrated lock body **800** includes a pair of extensions **802** at opposite ends **804**, **806** of the lock body **800**. The illustrated extensions **802** are substantially circular and may be shaped to interact with the blinds **320** formed in the wear sleeve retainer **204**. Accordingly, the lock body **800** may be aligned with the wear sleeve retainer **204** such that the extensions **802** interact with the blinds **320** to facilitate installation of the wear sleeve retainer **204**, for example, by using a tool to rotate the wear sleeve retainer **204**. The illustrated extensions **802** are positioned in a non-symmetrical arrangement in FIG. **8**, however, it should be appreciated that the respective locations of the extensions **800** may be particularly selected based on the blind configuration of the wear sleeve retainer **204**.

[0055] In various embodiments, the lock body **800** is configured to be adaptable to utilize existing tools, such as a ratchet wrench. Accordingly, the lock body **800** includes a coupling aperture **808** for receiving a mating tool part. In this manner, existing tools already present at the well site may be utilized with the lock body **800** to facilitate operations, thereby reducing clutter and leveraging existing components.

[0056] FIG. **9** is a front elevational view of the lock body **800**. In the illustrated embodiment, the extensions **802** have a length **900**, which may be particularly selected based on the size of the associated blinds **320**. In various embodiments, the extensions **802** may have different sizes, thereby providing a guide or indication regarding proper alignment or coupling to the wear sleeve retainer **204**. In this example, the non-symmetrical arrangement of the extensions **802** is further illustrated, in that the extensions **802** are not symmetrical about centerline **902**. This configuration may facilitate coupling at different circumferential positions of the wear sleeve retainer **204**, which may enable reduced force applications for installation and removal.

[0057] FIG. **10** is a side elevational view of the lock body **800**. As noted above, each of the extensions **802** extend for the same length **900**, but, in various embodiments the lengths **900**, along with the shapes of the extensions **802**, may be different. It should be appreciated that various other dimensions of the lock body **800**, such as the width, thickness, length, etc. may be particularly selected based on operating conditions.

[0058] FIGS. **11A-11E** illustrate perspective views of an installation procedure **1100** for securing the wear sleeve **514** within the bore **400** using the retainer system **200**. As noted above, various components and have been removed for simplicity with the following explanation, for example steps involving applying coatings, grease, lubrication, installing seals, and the like. Additionally, features such as threads and the like have been removed for clarity, however, threaded components have been described elsewhere herein. Furthermore, the steps may be performed in a different order, unless otherwise indicated. FIG. **11A** illustrates the wear sleeve **514** installed within the bore **400** such that the bore **400** engages the seal **518** positioned within the seal groove **520**. In various embodiments, the wear sleeve **514** is inserted into the bore **400**. FIG. **11B** illustrates the procedure **1100** of wear sleeve retainer **204** aligned with the bore **400**. As noted above, the wear sleeve retainer **204** may be installed within the bore **400** such that the mating threads **532** engage the threads **530** formed along the first bore section **502**.

[0059] FIG. **11C** illustrates the procedure **1100** of the installation of the wear sleeve retainer **204** using the lock body **800** and as associated tool **1102**, which in this instance is a ratchet wrench. As shown, the anti-rotation system **202** is already installed within the aperture **306**, for example, by bolting the locking mechanism **302** to the housing **102**, among other options. The locking fastener **304** of the illustrated embodiment is positioned through the locking mechanism **302** and backed off such that the locking fastener **304** does not engage the wear sleeve retainer **204**. Accordingly, the

wear sleeve retainer **204** may be installed within the bore **400** by rotating the wear sleeve retainer **204** to a predetermined position indicative of engagement with the threads **530**.

[0060] FIG. **11D** illustrates the procedure **1100** of the engagement of the wear sleeve retainer **204** via the locking fastener **304**. The locking fastener **304** extends through the locking mechanism **302** until it engages the valley **312**. In various embodiments, the angle **308** is particularly selected to engage the wear sleeve retainer **204** at a predetermined location. This embodiment illustrates engagement between 12 o'clock and 1 o'clock positions, however, different configurations may also be utilized in various embodiments. Furthermore, while a single anti-rotation system **202** is shown for the illustrated plunger assembly **104**, multiple anti-rotation systems **202** may be used, for example at different locations and/or the locking mechanism **302** may include multiple locking fasteners **304**.

[0061] FIG. **11E** illustrates the procedure **1100** of the packing nut **110** installed to engage the wear sleeve retainer **204**, for example via the threads **554**, **556** along with the plunger **106** installed within the plunger bore **402**. In various embodiments, the packing **524** is installed prior to installation of the packing nut **110**. It should be appreciated that components may be removed by reversing the steps described herein, for example, by removing the packing nut **110**, removing the packing **524**, disengaging the anti-rotation system **202**, removing the wear sleeve retainer **204**, and then removing the wear sleeve **514**.

[0062] FIG. **12A** is a perspective view of an embodiment of the anti-rotation system **202** using a ratchet and pawl system **1200**. In this embodiment, the locking mechanism **302** and locking fastener **304** are replaced with a pawl **1202** while the circumferential profile **310** (e.g., the valleys **312** and flats) of the wear sleeve retainer **204** function as a ratchet **1204**. It should be appreciated that, in various embodiments, the systems may be used interchangeably. By way of example, if there are 5 retainer systems **200** used on a fluid end **100**, one or more may use the locking mechanism **302** and the locking fastener **304** and one or more may use the pawl **1202**.

[0063] The illustrated pawl **1202** is coupled to the external face **300** of the fluid end **100**, for example via the aperture **306**, and is rotatable about a pawl axis **1206**. The pawl **1202** includes an inner profile **1208** and an outer profile **1210**. In this embodiment, the inner profile **1208** and outer profile **1210** are different, in that the inner profile **1208** has a more pronounced concave bend. It should be appreciated that inner and outer are used for illustrative and clarification purposes, and that such labels are not intended to limit embodiments of the present disclosure. For example, inner was selected in this instance because the inner profile **1208** is the leading edge of rotation about the pawl axis **1206** when moving the pawl **1202** into engagement with the ratchet **1204**.

[0064] The illustrated pawl **1202** includes a contact region **1212**, which is positioned to engage the valleys **312** of the wear sleeve retainer **204** (e.g., of the ratchet **1204**). In operation, the pawl **1202** is rotated about the pawl axis **1206**, in either a clockwise or counter-clockwise direction, to bring the contact region **1212** into the valleys **312**. The pawl **1202** may be secured, such as via a fastener or spring to block rotation in an opposite direction, and as a result, block rotation of the wear sleeve retainer **204**.

[0065] FIG. **12B** is a perspective view of the ratchet and pawl system **1200** where the pawl **1202** has been rotated about the pawl axis **1206** to engage the ratchet **1204**. Specifically, the contact region **1212** is positioned within the valley **312** after the pawl **1202** is rotated in the clockwise direction (compared to the position shown in FIG. **12A**). As noted above, further rotation of the pawl **1202** may be blocked, for example via a fastener, spring or the like, and as a result, the pawl **1202** blocks rotation of the wear sleeve retainer **204** in the counter-clockwise direction. In this manner, the wear sleeve retainer **204** may be secured to the fluid end **100** without externally bolting through the wear sleeve retainer, as is done with current systems. This simplifies installation and reduces the number of bolts utilized at the site. It should be appreciated that, in other embodiments, rotation directions may be changed and still be within the scope of the present disclosure.

[0066] FIG. **13** is a flow chart of an embodiment of a method **1300** for installing a retainer system.

It should be appreciated that this method, and all methods described herein, may include more or fewer steps. Additionally, the steps may be performed in a different order, or in parallel, unless otherwise specifically stated. The illustrated example includes installing the wear sleeve **1302**. For example, the wear sleeve may be arranged within a bore. In various embodiments, one or more landing or locating features may be included to facilitate arrangement of the wear sleeve. The wear sleeve retainer is installed **1304**. The wear sleeve retainer may be threaded to internal threads formed in a housing to eliminate external fasteners, which may be time consuming and difficult to install. An anti-rotation system may be utilized to block rotation of the wear sleeve retainer **1306**. For example, a fastener may apply an opposing force to the wear sleeve retainer to prevent rotation in a direction that would cause the wear sleeve retainer to back off. In various embodiments, internal components are installed **1308**, such as packing and the like. Then, a packing nut may be installed **1310**, among other components, to enable operation of the pumping system.

[0067] FIG. **14A** is a front view of an embodiment of the anti-rotation system **202** in which the locking fastener **304** extends through a portion of the housing **102**. In this example, the wear sleeve retainer **204** is illustrated installed within the housing **102** with the plunger **106** extending through the plunger bore **402**, as described above, such as in FIGS. **3** and **4**. The wear sleeve retainer **204** may be threaded into the housing **102**, as noted above, such that external fasteners are not used, or at least a reduced number of external fasteners are used. As such, the blinds **320** shown on the wear sleeve retainer **204** may be used with installation and removal, as noted herein.

[0068] In this example, the locking fastener **304** is positioned to extend through an aperture **1400** that extends through at least a portion of the housing **102**. This example shows the housing in phantom for convenience and to more clearly illustrate the position of the locking fastener **304**. For example, in at least one embodiment, the aperture **1400** is a counterbored opening that receives the locking fastener **304**, such as a threaded fastener, that extends through the opening to engage the wear sleeve retainer **204**. Threaded fasteners are provided as one example and are not intended to limit the scope of the present disclosure. Additional fasteners may include pins, tongue and groove fittings, bayonet connections, and the like. Furthermore, combinations of fasteners may be used and more than one fastener may be used to secure the wear sleeve retainer **204** into position. As shown, the locking fastener **304** engages the wear sleeve retainer along the profile **310**, such as at either valleys **312** (not shown) and/or flats **314** (not shown), as noted above. In at least one embodiment, engagement may be within the housing **102** (e.g., not visible from an external portion of the housing **102**) or may be outside of the housing **102**.

[0069] FIG. **14B** is a detailed perspective view of an embodiment of the anti-rotation system **202** in which a portion of the housing **102** is shown in phantom to illustrate interaction between the locking fastener **304** and the profile **310**. In this example, the locking fastener **304** extends through the aperture **1400** to be positioned in alignment with a portion of the profile **310** such that rotation applied to the profile **310** (e.g., to the wear sleeve retainer **304**) in a counter clockwise direction (from the perspective shown in FIG. **14B**), will drive the flat **314** (or a portion thereof) into contact with the locking fastener **304**, thereby preventing rotation of the wear sleeve retainer **304**. As shown, and described herein, the profile **310** includes the series of flats **314** and valleys **312**, where in this example the valleys **312** are not arcuate or curved like in FIG. **4**, but are illustrated as substantially planar and in a stepped configuration with respect to the flats **314**. As noted, a variety of different sizes for the flats **314** and valleys **312**, as well as shapes, may be provided for various embodiments within the scope of the present disclosure.

[0070] As noted herein, the configuration of the profile **310** may permit installation of the locking fastener **304** at a variety of different positions. That is, there is not a single location associated with the locking fastener **304** in this illustrated embodiment. Instead, the locking fastener **304** may be positioned to interact with any of the flat/valley **314**, **312** pairs in order to limit rotation of the wear sleeve retainer **204**. It should be appreciated that various other embodiments may have a particular location for installation of the locking fastener **304**.

[0071] The illustrated embodiment includes the aperture **1400** arranged proximate the port **546**, but such a position is for illustrative purposes and not intended to limit the scope of the present disclosure. For example, the aperture **1400** can be on an underside of the housing **102**, to the side, at an angle, or the like. Additionally, the aperture **1400** may be forwards of (e.g., closer to the face of the housing **102**) or rearwards of (e.g., farther from the face of the housing **102**) than the port **546**. The configuration of FIG. **14B** shows the angle **308** (not pictured, shown in FIG. **3**) to be approximately zero degrees (e.g., substantially perpendicular to the bore axis **402** (not pictured, shown in FIG. **4**)). This is in contrast to the angle **308** shown in FIGS. **3** and **4**, but it should be appreciated that the aperture **1400** may also be formed at a different angle **308** to facilitate installation of the locking fastener **304** at an angle other than 0 degrees. The angle of the aperture **1400** may be based, at least in part, on the profile **310** configuration to facilitate force transfer to block rotational movement of the wear sleeve retainer **204**.

[0072] In at least one embodiment, the profile **310** may be positioned axially inward, along the bore axis **402** (not pictured), compared to the threads **532**. That is, the profile **310** may be arranged further from the exterior of the housing **102**. However, in various other embodiments, the profile **310** may be positioned axially outward, along the bore axis **402**, compared to the threads **532**. It should be appreciated that the profile **310** may be arranged at various locations along the length of the wear sleeve retainer **304**. In at least one embodiment, the depth **316** may be particularly selected to position an interface for engagement between the locking fastener **304** and the profile **310** within the bore **400**. That is, the interaction may not be external to the housing **102**, as shown in FIG. **4**. However, it should be appreciated that an overhang or other component may be provided such that the engagement is external to the housing **102**.

[0073] FIG. **14C** is a cross-sectional view of an embodiment of the wear sleeve retainer **204** positioned within the bore **400** and secured into position by the locking fastener **304**. In this configuration, engagement is formed within the bore **400** such that the interaction between the locking fastener **304** and the wear sleeve retainer **204** is not visible from an exterior portion of the housing **102**. That is, the locking fastener **304** extends through the housing **102** to engage one or more portions of the wear sleeve retainer **204**, such as the profile **310**, which in this example may be arranged at an opposite end of the wear sleeve retainer when compared to the configurations of FIGS. **6** and **7**. In other words, the profile **310** in the illustrated configuration may be positioned axially away from the front face of the wear ring retainer **304**. It should be appreciated that the location of the profile **310** may vary based on one or more design considerations and that, in embodiments, the profile **310** may be closer to the front face than the position illustrated in FIG. **14C**.

[0074] As shown, the wear sleeve retainer **204** includes the valley **312** for receiving the locking fastener **304**. Accordingly, rotation of the wear ring retainer **204** will drive the associated flat **314** (not pictured) against the locking fastener **304**, thereby blocking and/or resisting the rotation. In this manner, the wear sleeve retainer **204** may still be threaded into the housing **102**, as noted above, to eliminate or reduce a number of external fasteners used to secure the wear sleeve retainer **204** to the housing **102**.

[0075] FIG. **15** is a perspective view of an embodiment of the anti-rotation system **204** in which the locking fastener **304** is shown in the form of a pad retainer **1500**. The pad retainer **1500** is positioned to bear against a front face **1502** of the wear sleeve retainer **204**, for example against a textured surface **1504**, to apply a frictional force to resist rotation of the wear sleeve retainer **204**. In this example, the wear sleeve retainer **204** is secured to the housing **102** via one or more internal threads, as noted above. During operation, rotational forces may be applied to the wear sleeve retainer **204**, which are undesirable for the reasons stated above. Accordingly, embodiments of the present disclosure may overcome those problems by providing the pad retainer **1500** to resist or block rotation of the wear sleeve retainer **204**. The pad retainer **1500** may be secured to the housing **102**, for example via one or more pad fasteners **1506**. The pad fasteners **1506** may be threaded

fasteners. As noted above, other wear sleeve retainers may include several fasteners to secure the wear sleeve retainers to an associated housing, but the illustrated embodiment includes two pad fasteners **1506**, which may correspond to the locking mechanism **302** shown in FIG. 3, thereby reducing the time and components used. It should be appreciated that the two pad fasteners **1506** are shown by way of example only and are not intended to limit the scope of the present disclosure. [0076] The textured surface **1504** may include knurling or one or more raised portions that interact with the pad retainer **1500** to generate friction, responsive to a rotational force applied to the wear sleeve retainer **304**, to block rotation of the wear sleeve retainer **304**. In at least one embodiment, the pad retainer **1500** may include a mating frictional surface (not visible) or may include a smooth surface, or some combination thereof. The pad retainer **1500** may be installed such that the pad retainer **1500** bears against the wear sleeve retainer **204**, which may provide a visual indication to the operator regarding installation. Furthermore, rotation may also be identified by visual review if scratches or other marks are shown on the front face **1502**, indicating rotation.

[0077] This application is a continuation of U.S. Non-Provisional application Ser. No. 18/771,874, filed Jul. 12, 2024, titled "PACKING BORE WEAR SLEEVE RETAINER SYSTEM," now U.S. Pat. No. 12,270,394, issued Apr. 8, 2025, which is a continuation of U.S. Non-Provisional application Ser. No. 17/890,975, filed Aug. 18, 2022, titled "PACKING BORE WEAR SLEEVE RETAINER SYSTEM," now U.S. Pat. No. 12,049,889, issued Jul. 30, 2024, which is a continuation-in-part of U.S. Non-Provisional application Ser. No. 16/916,593, filed Jun. 30, 2020, titled "PACKING BORE WEAR SLEEVE RETAINER SYSTEM," now U.S. Pat. No. 11,421,680, issued Aug. 23, 2022, the disclosures of which are incorporated herein by reference in their entireties for all purposes.

[0078] The foregoing disclosure and description of the disclosed embodiments is illustrative and explanatory of the embodiments of the disclosure. Various changes in the details of the illustrated embodiments can be made within the scope of the appended claims without departing from the true spirit of the disclosure. The embodiments of the present disclosure should only be limited by the following claims and their legal equivalents.

Claims

1. A wear sleeve retainer system, the system comprising: a wear sleeve retainer including: a body portion having a bore extending along an axis, a mating component extending from the body portion and into the bore, the mating component having a smaller diameter than a bore diameter of the bore of the body portion, and a profile positioned about the body portion, the profile including a plurality of receptacles; and a fastener positioned to engage one or more of the plurality of receptacles.
2. The wear sleeve retainer system of claim 1, wherein the body portion comprises a housing, and wherein the locking fastener extends through the housing having the bore.
3. The wear sleeve retainer system of claim 1, wherein the profile is positioned within the bore.
4. The wear sleeve retainer system of claim 1, wherein the profile includes sets of flats and valleys, and wherein respective flats of the sets of flats and valleys extend radially outward from respective valleys of the sets of flats and valleys.
5. The wear sleeve retainer system of claim 3, wherein the wear sleeve retainer further includes threads, wherein the threads comprise first threads, and the wear retainer system further comprising: second threads arranged along the bore extending from a face of the body portion to the mating component.
6. The wear sleeve retainer system of claim 5, wherein the locking fastener comprises one or more threaded fasteners.
7. The system of claim 3, wherein the wear sleeve retainer further includes threads, wherein the threads comprise first threads, and the wear retainer system further comprising: second threads

extending along at least a portion of the bore, the second threads ending prior to the mating component.

8. A fluid end comprising: a housing having a cavity and a bore extending toward the cavity; a wear sleeve positioned within the bore; a plunger bore extending in the wear sleeve; a plunger operative to reciprocate within the plunger bore; a wear sleeve retainer positioned to block axial movement of the wear sleeve; and an anti-rotation assembly to engage the wear sleeve retainer to block rotation of the wear sleeve retainer in one or more directions.

9. The fluid end of claim 8, wherein the anti-rotation assembly comprises: a locking fastener to engage the wear sleeve retainer, and wherein the locking fastener extends through the housing.

10. The fluid end of claim 8, wherein the anti-rotation assembly comprises: a pad retainer to be positioned to apply a force to a front face of the wear sleeve retainer.

11. The fluid end of claim 10, wherein one or more of the front face or the pad retainer includes a textured surface.

12. The fluid end of claim 8, wherein the wear sleeve retainer includes a profile positioned along an outer diameter to receive at least a portion of the anti-rotation assembly.

13. The fluid end of claim 12, wherein the profile is positioned within the bore.

14. The fluid end of claim 12, wherein the profile comprises a circumferential profile extending substantially around the wear sleeve retainer.

15. The fluid end of claim 8, wherein the wear sleeve retainer further comprises: at least a portion of the plunger bore, and a mating component extending into the plunger bore, the mating component having a smaller diameter than the plunger bore, the mating component also positioned to contact the wear sleeve, thereby to block axial movement of the wear sleeve.

16. The fluid end of claim 8, wherein the wear sleeve retainer further comprises: a plurality of blinds arranged circumferentially about the face, the plurality of blinds positioned radially inward of the profile.

17. The fluid end of claim 14, further comprising: a lock body having an extension, the extension corresponding to one or more blinds, the lock body adapted to connect to a rotational tool, thereby to drive rotation of the wear sleeve retainer about an axis.

18. A method for installing a retainer system, the method comprising: positioning packing seal elements within a diameter of a wear sleeve; positioning the wear sleeve within a bore located in a fluid end housing; securing a wear sleeve retainer to the fluid end housing, the securing including the step of positioning the wear sleeve retainer into the fluid end housing so that external portions of a body of the wear sleeve retainer engage internal portions in the bore; and securing an anti-rotation assembly to the wear sleeve retainer.

19. The method of claim 18, wherein the securing the anti-rotation assembly comprises: engaging an outer profile of the wear sleeve retainer via one or more locking fasteners.

20. The method of claim 19, wherein the securing the anti-rotation assembly comprises: engaging a front face of the wear sleeve retainer via a pad retainer, the pad retainer positioned to apply a frictional force to the wear sleeve retainer.
